

1. Administrative & Study Identification

Full Study Title: A Phase III Randomized Double-Blind Multi-Center Treat-Through Study to Evaluate the Pharmacokinetics, Safety and Efficacy of Induction and Maintenance Therapy With Afimkibart (R07790121) in Children Aged 2 - 17 Years With Moderately to Severely Active Ulcerative Colitis **Short Title/ Acronym:** AMETRINE-PEDS **Protocol Number:** CA45905 (EU CTIS: 2025-522518-22-00) **NCT Number:** NCT07158242 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Afimkibart (R07790121 / PF-06480605 / RVT-3101) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of afimkibart induction and maintenance therapy in achieving clinical remission in pediatric participants (aged 2 to 17 years) with moderately to severely active ulcerative colitis. **Secondary Objectives:** To assess the pharmacokinetics (PK), safety, tolerability, endoscopic improvement, and overall clinical response of afimkibart in this pediatric demographic.

2.2 Background & Rationale To address the unmet need for effective targeted therapies in pediatric patients with moderately to severely active Ulcerative Colitis (UC). Despite current standards of care, many children demonstrate intolerance or an inadequate response to conventional treatments (systemic corticosteroids, immunomodulators, and/or existing biologic therapies). Afimkibart, a novel TL1A inhibitor, targets inflammation and fibrosis pathways, offering a promising alternative for inducing and maintaining remission in this population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled / Active Dose Comparator **Blinding:** Double-blind (Quadruple Masking) **Randomization:** Randomized, Parallel Assignment

3.2 Planned Enrollment & Duration Target \$N\$: 100 participants **Number of Sites:** Multi-center (Global) **Estimated Study Start:** 31-Mar-2026 **Estimated Primary Completion:** 31-Mar-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Children aged 2 to ≤ 17 years with a bodyweight ≥ 10 kg. 2. Confirmed diagnosis of moderately to severely active Ulcerative Colitis (UC). 3. Demonstrated intolerance or inadequate response (IR) to one or more of the following categories of drugs: systemic corticosteroids, immunomodulators, and/or biologic therapies as outlined in the protocol.

4.2 Exclusion Criteria 1. Monogenic disorder pertaining to infant-onset inflammatory bowel disease (IBD). 2. Current diagnosis of Crohn's disease (CD), abdominal/intrabdominal/perianal fistula and/or abscess, indeterminant colitis, IBD-unclassified, microscopic colitis, ischemic colitis, infectious colitis, radiation colitis, or active diverticular disease. 3. Presence of an ostomy or ileoanal pouch; current diagnosis or suspicion of primary sclerosing cholangitis; or any major surgery within 6 weeks prior to screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants will receive afimkibart induction therapy followed by maintenance dosing intervals. **Route of Administration:** Intravenous (IV) infusion for induction, followed by Subcutaneous (SC) injection for maintenance. **Duration of Treatment:** Treat-through design (evaluating both short-term induction and long-term maintenance).

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of baseline UC therapies (e.g., aminosalicylates) as permitted per study protocol. **Prohibited:** Concurrent use of other unapproved biologic therapies, investigational targeted agents, or new immunosuppressive regimens during the blinded trial period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent/ Assent, Medical History, Baseline Endoscopies, Clinical Scoring
Baseline	Day 0	Randomization, First IV Infusion (Induction), PK/Safety Labs	Interim Induction & Maintenance Onsite Visits: IV/SC Dosing, Efficacy Assessments, Endoscopic Reviews, Safety Labs End of Study Study Completion Final Clinical/Endoscopic Evaluation, SC Drug Accountability, Safety Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of participants achieving Clinical Remission at the end of the evaluation period. **Measurement Method:** Evaluated via standardized UC clinical scoring systems (e.g., modified Mayo score incorporating stool frequency, rectal bleeding, and endoscopic subscores).

7.2 Secondary & Exploratory Endpoints * Endoscopic improvement/remission rates. * Incidence and severity of treatment-emergent Adverse Events (AEs). * Pharmacokinetic (PK) profile parameters of Afimkibart (R07790121) in the pediatric population. * Changes in inflammatory biomarkers (e.g., fecal calprotectin, CRP).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical exams, lab assessments, caregiver diaries, and onsite clinical reviews. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (iDMC) will oversee unblinded safety and efficacy data throughout the trial's duration.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy endpoints; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target N=100\$ powered to adequately assess pharmacokinetics and detect a clinically meaningful difference in clinical remission rates between the investigational arm(s) and placebo. **Handling of Missing Data:** Non-Responder Imputation (NRI) utilized for binary clinical efficacy endpoints (e.g., clinical remission); mixed-model repeated measures (MMRM) for continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Utilizing fully validated Electronic Data Capture (EDC) systems with detailed eCRF locking mechanisms and data clarification audit trails.

11. Study Identifiers & Governance

Primary ID: CA45905 **Registry Number:** NCT07158242 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** AMETRINE-PEDS **Official Regulatory Title:** A Phase III Randomized Double-Blind Multi-Center Treat-Through Study to Evaluate the Pharmacokinetics, Safety and Efficacy of Induction and Maintenance Therapy With Afimkibart (R07790121) in Children Aged 2 - 17 Years With Moderately to Severely Active Ulcerative Colitis

12. Clinical Framework (Ulcerative Colitis Specific)

Primary Condition: Moderately to Severely Active Ulcerative Colitis (Pediatric) **Diagnosis Threshold:** Endoscopic and clinical confirmation of moderate to severe active disease **Medical Methodology:** Biologic Induction and Maintenance Therapy (TL1A Inhibition) **Optimization Target:** Clinical Remission and Endoscopic Mucosal Healing

13. Study Procedures & Methodology

Management Pathway: Double-blind treat-through clinical pathway for UC **Education Protocol (HCP):** Protocol-specific training for pediatric UC assessment (Mayo scoring) and IV/SC administration protocols **Education Protocol (Patient):** Caregiver/Patient education on SC injection techniques, symptom diary management, and AE reporting **Quality Audit Mechanism:** Caregiver eDiary completion rates, PK sampling adherence, and SC drug accountability tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to study primary completion (expected spanning multi-year follow-up for maintenance evaluation) **Onsite Visit Cadence:** Frequent visits during the IV induction phase, shifting to regular 4-to-8 week intervals during the SC maintenance phase **Remote Monitoring Frequency:** Telehealth and eDiary reviews every 2 to 4 weeks (Q2W-Q4W) **Checklist Review Interval:** Regular clinical/endoscopic scoring review intervals aligned with primary endpoint timeframes

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				